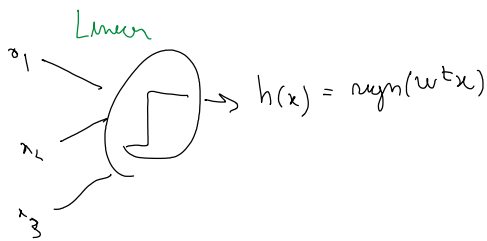


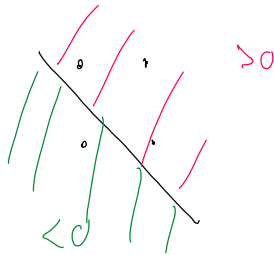
Aula 6

Wednesday, March 18, 2026 8:01 AM

Perceptron e Neural Networks



perceptron com neurônio linear



Exemplos

$$1) \quad x_1 = (1, 0, 0) \\ w = (-0.5 \ 1 \ 1) \quad \text{sign}(w \cdot x^{(1)}) = -1 \quad y^{(1)} = 1 \\ w = (-0.5 \ 1 \ 1) + 1 \cdot (1, 0, 0) = (0.5 \ 1 \ 1)$$

$$2) \quad x^{(2)} = (1, 0, 1) \quad \text{sign}(w \cdot x^{(2)}) = 1 \quad y^{(2)} = 1 \\ w = (0.5, 1, 1) \quad \text{sign}(w \cdot x^{(3)}) = 1 \quad y^{(3)} = 1 \\ x^{(3)} = (1, 1, 0) \quad \text{sign}(w \cdot x^{(4)}) = 1 \quad y^{(4)} = -1 \\ x^{(4)} = (1, 1, 1)$$

$$w = (0.5 \ 1 \ 1) + (-1) \cdot (1 \ 1 \ 1) = (-0.5 \ 0 \ 0)$$

$$3) \quad \text{sign}(w \cdot x^{(1)}) = -1 \quad y^{(1)} = 1 \\ w = (-0.5 \ 0 \ 0) \\ w = (-0.5 \ 0 \ 0) + 1 \cdot (1, 0, 1) = (0.5, 0, 1)$$

$$4) \quad \text{sign}(w \cdot x^{(2)}) = 1 \quad y^{(2)} = 1 \\ w = (0.5 \ 0 \ 1) \\ \text{sign}(w \cdot x^{(1)}) = 1 \\ \text{sign}(w \cdot x^{(3)}) = 1$$

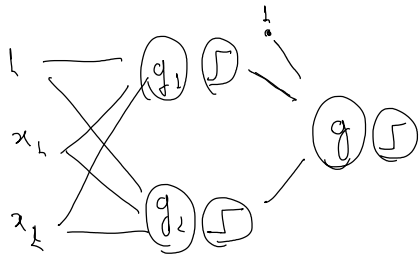
Perceptron Full

x_1	x_2	x_3	y
0	0	0	0
0	1	1	1

$$\begin{array}{cc|c} 1 & 0 & 1 \\ 1 & 1 & 0 \end{array}$$

cons comment
2 perceptors

Perceptron Network for XOR



Neural Network

x_1 ,
 x_2 ,
 x_3 ,
 x_4 .

